

IA model flexibility and parameter constraints

NLA: restricted response

Few nuisance parameters
Restricted response flexibility
Bias if the model is inadequate

TATT: more response terms

Broader tidal responses
More adjustable parameters
Can be harder to constrain

Compact reconstruction of a flexible response family

Experiment in this report

13 sampled
shape parameters

$\mathcal{A}_\Theta(k, z)$
 31×101 grid

Log-space PCA
or autoencoder

Matched validation
errors

Reconstructed
response

Two latent
coordinates

Further tests before cosmological use

Errors in lensing predictions
Priors for latent coordinates; unbiased inference